

PANITHAN LERTSUNTIVIT

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SUMMARY & SKILLS

UC Berkeley Mechanical Engineering Graduate and US Citizen with proven experience in instructing engineering courses focused on hardware integration, microprocessor architecture, and embedded development. Hands-on experience in manufacturing, rapid prototyping, electromechanical integration, and validation testing. A proactive problem-solver eager to apply hands-on technical skills to build next-generation platforms.

Software & Analysis: SolidWorks, Python, AutoCAD, Onshape, ESP-IDF, C, LabVIEW, MATLAB

Hardware & Manufacturing: NI DAQ, System Integration, Hardware Validation, Oscilloscopes, 3D Printing, GD&T, Machining (Mill/Lathe), Wire Harnessing, Cleanroom Protocols

EDUCATION

University of California, Berkeley

Graduated December 2025

B.S. in Mechanical Engineering

GPA: 3.73

Irvine Valley College

Graduated May 2023

A.S. in Engineering

GPA: 4.00

PROFESSIONAL EXPERIENCE

UC Berkeley Mechanical Engineering

Berkeley, CA

Teaching Assistant | ME 103 - Experimentation and Measurements *August 2025 - December 2025*

- Guided 160+ students in validating sensor data using NI USB-6211 DAQs and LabVIEW, debugging DAQmx configurations and teaching oscilloscope proficiency to resolve data acquisition failures.
- Mentored student teams on instrumenting dynamic systems (e.g. using strain gauges, LVDTs, etc.) and evaluated measurement fidelity for final technical presentations and reports.
- Managed engineering lab resources by implementing a centralized scheduling system for high-value equipment (Instron, Wind Tunnel, IC Engine, Camshaft Dynamics, Thermal Camera), streamlining coordination with shop staff to ensure mandatory supervision and safety compliance.

Teaching Assistant | ME 100 - Electronics for the Internet of Things

August 2024 - May 2025

- Mentored 25+ groups in troubleshooting software and hardware failures, ranging from algorithms implementation to circuit continuity, enabling the development of ESP32-based IoT prototypes.
- Delivered weekly reviews on circuit analysis (KCL/KVL, Op-Amps, Transistors) and communication protocols, creating high-fidelity visual resources and recordings used by the 200+ student cohort.

National Tsing Hua University

Hsinchu, Taiwan

Research Intern | Yang Lab for Biomechanics

June 2025 - August 2025

- Engineered the mechanical architecture for a bio-inspired fluid dynamics test rig to simulate cuttlefish ink flow patterns, designing a custom linear actuation system (stepper motors, lead screws) to strictly control variable volumetric flow rates of ink and mucus.
- Accelerated the project timeline by delivering a validated SolidWorks assembly and comprehensive Bill of Materials (BOM), utilizing Motion Studies to verify mechanism travel limits and sourcing compatible components from industrial vendors (e.g., Misumi).

- Provided 1-on-1 technical instruction in Calculus I/II, Physics, Chemistry, and Engineering Statics, deconstructing complex topics and help long-term students achieve academic excellence.
- Mentored students in Computer-Aided Design, guiding them through interface navigation, modeling complex 3D geometries, and creating standardized engineering drawings.
- Optimized tutoring center operations by managing scheduling logistics and mapping software availability across workstations to improve student access to technical resources.

ORGANIZATIONS

Aerospace SAE at UC Berkeley | *Director of Avionics*

2024 - 2025

- Led the integration of avionic subsystem for the 2025 SAE Aero Design West competition, ensuring compliance with the 750W power limit and contributing to a 3rd Place National finish.
- Engineered a ESP32-based thrust test rig to characterize propulsion data across multiple configurations; analyzed results and optimized the propulsion system that maximized thrust.
- Designed and fabricated custom wire harnesses, implementing Molex Mega-Fit connectors to ensure mechanical integrity and electrical continuity under high-vibration flight conditions.

Thai American Cultural Association at UC Berkeley | *Finance Chair*

2025

- Managed and audited financial transactions for club initiatives.

PROJECTS & INTERESTS

eV/STOL – Developed a tilt-rotor tricopter, utilizing OnShape and Arduino-based flight controllers to achieve hybrid vertical/horizontal flight capabilities.

Microfabrication of Transistors – Fabricated 4-mask NMOS devices in a cleanroom environment; performed oxidation, photolithography, diffusion, metalization, and metrology validation.

Effects of Filament Dyes – Conducted tensile and bending test on 3D printed specimens using an Instron 5500, analyzing data to quantify the mechanical impact of pigment additives in PLA filament.

Real-Time Plotter – Developed a Wi-Fi-enabled plotting system utilizing LabVIEW for telemetry and an ESP32 for low-level stepper motor control via TCP/IP protocols.

Mobile Desk – Prototyped a portable workstation conforming to ASME Y14.5M GD&T standards; fabricated aluminum components using manual machining to +/- 0.001" tolerances.

Wall Climbing Vehicle – Designed a suction-based rover using SolidWorks, utilizing laser-cut living hinges to achieve complex flexible geometries for rapid chassis prototyping.

Dark Matter Star Model – Developed a Dark Matter self interaction model by numerically solving the Tolman-Oppenheimer-Volkoff equations; presented at the 2023 UCI Building Bridges Conference.

Hobbies: Exploring Cultural Landmarks and Asian Gourmet Cooking